Deep Learning in Finance

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Course Description

This course consists of seven sessions, each lasting 2.5 hours, and follows the format of an advanced topics seminar. It provides an in-depth exploration of advanced deep learning techniques with applications to financial time series data.

Prior familiarity with deep neural networks and backpropagation derivation is required. The course begins with an overview of the theoretical foundations that justify deep learning models, including the virtue of complexity in return prediction as the finance-facing instance of the overparameterization story. It then delves into topics such as Transformers, Neural ODEs and neural point processes, GANs, Diffusion Models, and Graph Neural Networks (GNNs), with each architecture paired with the asset pricing literature that uses it. The course culminates in a discussion of Bayesian deep networks for factor modeling, where uncertainty quantification is brought to bear on deep factor models.

Evaluation

- **Research Paper Presentation**: 40%
- **Final Project**: 60%
- **Home works**: optional, not graded

Weekly homeworks are provided as practice and are strongly recommended as preparation for the final project, but they are not collected or graded.

Materials

- **Lecture slides**
- **Companion GitHub Repository**: Python/PyTorch

Textbooks

Primary Text

Chris Bishop, Hugh Bishop (2023) *Deep Learning*, The MIT Press.

Optional Texts

- Aston Zhang, Zachary C. Lipton, Mu Li, Alexander J. Smola (2023), *Dive into Deep Learning*, Cambridge University Press.
- Simon Prince, (2023) *Understanding Deep Learning*, The MIT Press.
- Jakub M. Tomczak (2022) *Deep Generative Modeling*, Springer.
- Ovidiu Calin, (2020) *Deep Learning Architectures*, Springer.
- Daniel A. Roberts, Sho Yaida, Boris Hanin (2022) *The Principles of Deep Learning Theory: An Effective Theory Approach to Understanding Neural Networks*, Cambridge University Press.
- Jong Chul Ye (2022) *Geometry of Deep Learning: A Signal Processing Perspective*, Springer.
- Agostino Capponi, Charles-Albert Lehalle (eds.) (2023) *Machine Learning and Data Sciences for Financial Markets: A Guide to Contemporary Practices*, Cambridge University Press.

Lectures

- **Lecture 1: Deep Learning - Theoretical Justifications**
  - **Scope:** This lecture provides a high-level theoretical perspective on why deep learning works, focusing on approximation, optimization, and generalization. Topics include the universal approximation theorem, double descent regimes, and overparameterization. The lecture emphasizes the current limitations in theoretical understanding that explain the empirical success of deep learning models. We also explore semi-empirical and practical methodology to check the generalizability of deep learning models. The lecture closes with the *virtue of complexity* literature, which carries the benign overfitting and double descent arguments into return prediction and shows that the ridgeless limit of a heavily overparameterized forecaster can dominate its parsimonious counterpart out of sample.
  - **Readings:**
    - Belkin, M., Hsu, D., Ma, S., and Mandal, S. (2019) Reconciling Modern Machine Learning and the Bias-Variance Tradeoff. Proceedings of the National Academy of Sciences (PNAS), 116(32), 15849–15854.
    - Tomaso Poggio, Andrzej Banburski, Qianli Liao (2019) Theoretical Issues in Deep Networks: Approximation, Optimization and Generalization Proceedings of the National Academy of Sciences (PNAS), 117 (48) 30039-30045.

    - Yasaman Bahria, Ethan Dyera, Jared Kaplanb, Jaehoon , (2024) Explaining neural scaling laws Proceedings of the National Academy of Sciences (PNAS), 117 (48) 30039-30045.
    - Charles Martin and Michael Mahoney (2021), Implicit self-regularization in deep neural networks
    - Oualid Missaoui (2025), Inductive Triplet Fine Tuning for Small Language Models
    - Bryan Kelly et al. (2023) The Virtue of Complexity in Return Prediction
    - Bryan Kelly et al. (2025) Understanding The Virtue of Complexity
    - Bryan Kelly et al. (2024) Large (and Deep) Factor Models

- **Lecture 2: Transformers for Financial Time Series Forecasting**
  - **Scope:** This lecture introduces the Transformers architecture and discusses its advantages over recurrent based sequence to sequence architectures. It also illustrates its application to financial time series forecasting. The lecture then turns to deep learning statistical arbitrage, where attention and convolutional sequence models are used to map the residual time series of an arbitrage portfolio into a trading policy, as a full-scale application of sequence modeling to a cross-sectional trading problem.
  - **Readings:**
    - Chapter 12, Transformers, from the primary textbook.
    - Vaswani et al, (2017) Attention is All you Need NIPS.
    - Edmond Lezmi, Jiali Xu, (2023) Time Series Forecasting with Transformer Models and Application to Asset Management, Amundi Institute.
    - Jorge Guijarro-Ordonez et al. (2022), Deep Learning Statistical Arbitrage

- **Lecture 3: Neural Ordinary Differential Equations (NODE) and Neural Point Processes for Financial Time Series**
  - **Scope:** This lecture introduces Neural ODEs and its formulation and how they extend the concept of ResNets to continuous layers. It also provides and explanation of the architecture and implementation of NODEs in practice, and illustrates its application to financial time series forecasting. The second half extends the continuous-time latent state to event data: the *neural Hawkes process* replaces the fixed exponential kernel of a classical Hawkes process with a continuous-time LSTM, and neural jump SDEs place the same idea directly on top of the Neural ODE machinery. We apply the resulting conditional intensity to a hazard problem in quantitative finance, modeling the arrival intensity of credit events and of self-exciting order

flow.

- **Readings:**
  - Chapter 1 and 2 from the PhD dissertatin of Patrick Kidger, On Neural Differential Equations, 2022.
  - Ricky T.Q. Chen, Yulia Rubanova, Jesse Bettencourt, David Duvenaud, (2018) Neural Ordinary Differential Equations, NIPS
  - Jingsui Li; Wei Zhu; Zhang Chen; Chao Pei (2023) Financial Time Series Prediction via Neural Ordinary Differential Equations Approach International Annual Conference on Complex Systems and Intelligent Science
  - Hongyuan Mei, Jason Eisner (2017) The Neural Hawkes Process: A Neurally Self-Modulating Multivariate Point Process, NIPS
  - Junteng Jia, Austin R. Benson (2019) Neural Jump Stochastic Differential Equations, NeurIPS
  - Emmanuel Bacry, Iacopo Mastromatteo, Jean-Francois Muzy (2015) Hawkes processes in finance, Market Microstructure and Liquidity, 1(1).
  - Justin Sirignano, Apaar Sadhwani, Kay Giesecke (2021) Deep Learning for Mortgage Risk, Journal of Financial Econometrics, 19(2), 313–368.

- **Lecture 4: Generative Adversarial Models (GANs) and Adversarial Estimation in Finance**
  - **Scope:** This lecture introduces GANs and presents an overview of its components and architecture. It also derives the underlying updating equations for its training. It then illustrates the applications of GANs in financial synthetic data generation. Finally, the lecture shows that the adversarial minimax structure is not confined to sample generation: in deep learning for asset pricing, one network parameterizes the stochastic discount factor while an adversarial network searches over the moment conditions that the SDF fails to price, which is exactly the GAN objective applied to a no-arbitrage estimation problem.
  - **Readings:**
    - Chapter 17, Generative Adversarial Models, from the primary textbook.
    - Ian J. Goodfellow et al. (2014) Generative Adversarial Nets
    - Adil Rengin Cetingoz, and Charles Albert Lehalle, (2025) Synthetic Data for Portfolios: A Throw of the Dice Will Never Abolish ChancearXiv, arXiv:2501.03993
    - Luyang Chen et al. (2021) Deep Learning in Asset Pricing

- **Lecture 5: Generative diffusion Modeling for Financial Synthetic Data**

- **Scope:** This lecture introduces the generative diffusion models, provides and overview of its primary components and typical architecture, and derive the underlying equations for training. In addition, it illustrates its application to synthetic financial data generation.
- **Readings:**
  * Chapter 20, Diffusion Models, from the primary textbook.
  * Stanley Chan , (2025) Tutorial on Diffusion Models for Imaging and Vision
  * Andrew Lesniewski and Giulio Trigila (2025) Beyond Monte Carlo: Harnessing Diffusion Models to Simulate Financial Market Dynamics
  * Oualid Missaoui (2025) Deep Generative Modeling for Covariance Denoising in Finance
  * Zhuohan Wang, Carmine Ventre (2024) A Financial Time Series Denoiser Based on Diffusion Models, ICAIF

- **Lecture 6: Graph Neural Networks for Financial Data**
  - **Scope:** This lecture will introduce Graph Neural Networks (GNNs) and explain GNN architecture as well as graph embeddings. It will also illustrate how can GNN be potentially leveraged to model interconnected assets or networks of trading relationships.
  - **Readings:**
    * Chapter 13, Graph Neural Networks, from the primary textbook.
    * Franco Scarselli; Marco Gori; Ah Chung Tsoi; Markus Hagenbuchner; Gabriele Monfardini (2009), The Graph Neural Network Model IEEE Transactions on Neural Networks
    * Gregoire Pacreau, Edmond Lezmi, Jiali Xu, (2022) Graph Neural Networks for Asset Management

- **Lecture 7: Bayesian Deep Networks for Factor Modeling**
  - **Scope:** This lecture closes the course by putting a posterior on the weights. We review why point-estimated deep factor models are hard to trust in production — no interval estimates, no principled ranking of factor importance, no account of model risk — and then develop the Bayesian treatment: priors over weights, variational inference and Bayes by backprop, MC dropout as an approximate posterior, and sampling-based alternatives. The lecture follows Dixon and Polson's Bayesian deep fundamental factor model, which yields a predictive distribution of excess returns together with uncertainty quantification, ranking of factor importances, and estimation of interaction effects, and contrasts it with the deep factor models seen earlier in the course.
  - **Readings:**

* Chapter 14, Sampling, from the primary textbook.
* Matthew Dixon, Nicholas Polson (2023) Bayesian Deep Fundamental Factor Models, Chapter 32 in A. Capponi and C.-A. Lehalle (eds.), *Machine Learning and Data Sciences for Financial Markets*, Cambridge University Press.
* Matthew F. Dixon, Nicholas G. Polson (2020) Deep Fundamental Factor Models, SIAM Journal on Financial Mathematics.
* Charles Blundell et al. (2015) Weight Uncertainty in Neural Networks, ICML
* Yarin Gal, Zoubin Ghahramani (2016) Dropout as a Bayesian Approximation: Representing Model Uncertainty in Deep Learning, ICML
* Laurent Valentin Jospin et al. (2022) Hands-on Bayesian Neural Networks — a Tutorial for Deep Learning Users, IEEE Computational Intelligence Magazine
* Guanhao Feng et al. (2023) Deep Learning in Characteristics-Sorted Factor Models
* Guanhao Feng et al. (2018) Deep Learning Factor Alpha